

scripts/autonomous-qa/run-iteration.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)

Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.J / §6.L (anti-bluff), §6.AK (device-gate auth), §6.H (credential handling) **Classification:** project-specific

Overview

Runs **one QA iteration** against an already-booted emulator + already-up backend (the matrix cell executed by run-matrix.sh):

1. **FRESH client install** (uninstall + install -r) — clean onboarding state
2. Start recording: cleared logcat + chunked screenrecord (180 s cap/chunk)
3. Run Challenge70AutonomousQaProviderMatrixTest parameterized via gradle connectedDebugAndroidTest instrumentation arguments
4. Stop recording (+ a best-effort final-screen.png still while the app is up)
5. Parse the JUnit verdict → curated verdict.json

Raw media/logs go to <evidence-dir>/raw/ (gitignored); verdict.json + junit.xml are curated.

Usage

```
scripts/autonomous-qa/run-iteration.sh \
  --backend goapi|apiapp --providers <csv> --query <str> \
  --serial <adb-serial> --api-url <url> --evidence-dir <dir>
```

All of --backend, --providers, --query, --serial, --evidence-dir are required; --api-url defaults to https://127.0.0.1:8443.

The goapi auth key (§6.AK / §6.H)

For the goapi backend the client must authenticate /v1 with the Go API's configured Lava-Auth key. The script derives it from the same root .env the compose stack feeds the container

(LAVA_AUTH_ACTIVE_CLIENTS: name:uuid) — it needs the UUID only, NOT the HMAC secret. The derived key is credential-equivalent and is **never echoed**. Without an entry the iteration runs keyless (and /v1 will 401 — recorded honestly).

Verdict decision table (anti-bluff)

Challenge70 logs C70-RESULT ... DOWNLOAD-OK only after the user-visible download/magnet affordance is confirmed on screen — hard evidence the real user flow reached the download. Separately, the test's MainActivity teardown can crash with the known-open upstream defect **LVA-008** (a process-FATAL IllegalStateException at activity destroy) that fails the JUnit run without being a product defect. Signals grepped from logcat + JUnit XML:

- marker_download_ok — the on-screen download affordance was confirmed
- teardown_known_lva008 — logcat/JUnit contains BOTH LVA-008 phrases
- other_failure_signal — ANY failure unrelated to LVA-008 (explicit C70 download-step failure marker, or any non-LVA-008 AssertionError)

Condition	Verdict
gradle_rc=0, 0 failures/errors, tests>0, not all skipped	PASS
gradle_rc=0 but tests=0	FAIL — a green run with zero executed tests is a bluff by construction
all tests skipped	SKIP
marker AND lva008-teardown AND no other failure signal	PASS (PASS-via-marker; all three MUST hold)
anything else	FAIL

A missing client APK writes a FAIL verdict.json immediately (never leaves a stale PASS behind — the 2026-07-03 incident).

Exit codes

- 0 — verdict PASS
- 1 — verdict FAIL/SKIP
- 2 — usage error or setup error (e.g. missing APK; a FAIL verdict is written)

verdict.json fields

backend, providers, query, serial, gradle_rc, tests, failures, errors, skipped, marker_download_ok, teardown_known_lva008, other_failure_signal, verdict, note, junit_xml, raw_dir.

Companion files

- scripts/autonomous-qa/run-matrix.sh — the caller
- scripts/autonomous-qa/lib-summary.sh — parses these verdict files back out
- scripts/autonomous-qa/vision-analyze.sh — analyzes the raw/recordings